

PHISHING WEBSITE DETECTION USING MACHINE LEARNING

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ABSTRACT: Phishing is a common tactic employed to deceive unsuspecting individuals into revealing their personal information through fraudulent websites. These phishing websites often steal passwords, usernames, and sensitive data related to online financial transactions. Phishers create websites that mimic the appearance and language of legitimate web pages to trick their targets. To combat the evolving techniques used in phishing attacks, it is crucial to implement anti-phishing measures that can detect such attempts. Machine learning can play a significant role in stopping phishing attacks. This proposal presents a compilation of features and machine learning-based algorithms for identifying phishing attempts. Attackers frequently resort to phishing because it is easier to trick victims into clicking on seemingly legitimate yet malicious links, compared to circumventing computer security protocols. By replicating the official website's logo and other elements, the malicious link in the message body gives the impression of being affiliated with the legitimate website. This proposal explores the distinguishing characteristics of phishing domains, also known as scam domains, and emphasizes the importance of identifying them. Additionally, it discusses machine learning and natural language processing techniques that can be employed to achieve this goal.

Keywords: Phishing, Personal Information, Machine Learning, Malicious Links, Phishing Domain Characteristics.

I. INTRODUCTION

Phishing websites are a serious threat to people and businesses because they target their confidential data and financial resources. These bogus websites are made to trick people into giving up their personal information, including passwords, credit card information, and social security numbers. In order to ensure online security, it is now essential to identify and block phishing websites.

Attacks that use phishing have developed throughout large datasets with attributes taken from both legal and phishing websites can be used to train machine learning algorithms. These qualities could be found in URL traits, website content, SSL certificates, IP reputation, and other places. Machine learning models can successfully distinguish between reliable and phishing websites by observing patterns and spotting signs of criminal intent.

Techniques for natural language processing (NLP) are essential for detecting phishing websites. Phishing attempts frequently employ misleading emails and messages to try to persuade recipients to visit malicious websites. These communications' textual content can be analyzed by NLP algorithms, which can spot patterns, grammatical mistakes, or dubious wording that point to a phishing attempt. The prevention of financial losses and reputational damage

to organizations is greatly enhanced by the identification of phishing websites. Security systems may proactively identify and prevent phishing websites using machine learning and NLP techniques, adding another line of defense against these nefarious actions.

It will examine numerous methods, algorithms, and attributes that can be used to identify phishing websites in this study. We will talk about the merits and drawbacks of various approaches as well as prospective methods for raising their precision. We can strengthen internet security and shield people from falling for these fraudulent schemes by staying ahead of the constantly changing strategies employed by phishing attackers.

Security specialists are currently most concerned about phishing since it's so simple to develop a phony website that closely resembles a genuine one. Not all users and some phishing victims are experts at spotting malicious websites. The main objective of the attacker is to acquire bank account credentials [1-3]. Because users are not aware of them, phishing assaults are more successful. It is difficult to combat phishing attempts that prey on user vulnerabilities, so it is crucial to advance phishing detection techniques. Phishing is a common kind of extortion in which hostile websites pretend to be trustworthy organizations in order to get personal data like passwords, account logins, or MasterCard numbers [4-6]. Phishing is creating a brand-new hybrid strategy nevertheless, one that goes beyond frameworks and open programming.

Phishing is a type of scam that employs social engineering tactics to assume the identity of another individual and disseminate private or sensitive information, such as passwords or open credentials or the online communications industry. Phishing exploits links on phishing websites to deceive people into going to a fake website. Utilize spoof messages that appear to come from reliable sources, such financial institutions and online business objectives, and are endorsed by them.

II. LITERATURE SURVEY

H. Huang et al. (2009) suggested a methodology to distinguish legitimate websites from phishing websites utilizing page section similarities to decompose universal resource locator tokens in order to offer prediction accuracy. By examining log data from the real website server, S. Marshall et al. (2017) developed a method to recognise phishing websites. Software made to recognise phishing websites or off-hook software Excellent attributes displayed by free include superb accuracy, total independence, and lovely language freedom. Quick selection, dynamic

phishing, and the capacity for phishing technique evolution [7 - 10].

A summary of the major machine learning methods for phishing website identification is given in this review study. The article compares the effectiveness of various machine learning algorithms in identifying phishing websites and explores feature extraction techniques such URL-based, content-based, and SSL certificate-based features. The report also emphasizes the restrictions and difficulties in this area.

The effectiveness of different machine learning methods, such as decision trees, random forests, support vector machines, and neural networks, for phishing website identification is compared in this study. Utilising several feature sets, it assesses their accuracy, precision, recall, and F1-score. The results reveal the efficiency of various algorithms and characteristics in phishing website detection.

This study suggests a machine learning-based method for identifying phishing websites. It makes use of website-extracted features such URL-based features, textual features, and graphic features. Using real-world phishing datasets, the study applies various machine learning techniques, such as logistic regression, k-nearest neighbors, and gradient boosting, and assesses their performance. An efficient machine learning method for identifying phishing websites is shown in this paper. Machine learning models are trained using a combination of URL-based, content-based, and website reputation-based elements. The study examines the effectiveness of several algorithms, including support vector machines, random forests, and k-nearest neighbors, and talks about how feature selection affects the accuracy of detection.

III. PROPOSED METHODOLOGY

The detection of phishing websites is a crucial step in the fight against online fraud and identity theft. Machine learning methods have become effective tools for spotting and reducing the hazards brought on by phishing attempts. We can automate the process of finding and blocking phishing websites by utilizing the powers of machine learning, offering a proactive defense against these harmful actions as shown in figure 1.

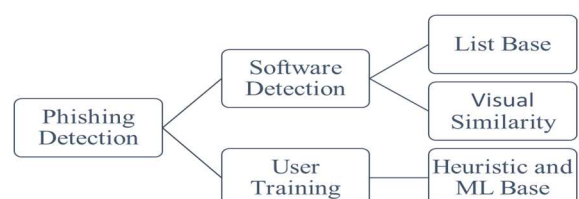


Fig 1: Phishing Detection using Machine Learning

Machine learning algorithms can be trained to examine a range of information extracted from websites to assess their propensity to be phishing sites. Key characteristics that can be exploited for phishing website detection include:

URL analysis: Suspicious patterns, such as misspelled domain names, extra subdomains, or long and winding pathways, are frequently seen in phishing URLs. To identify these patterns and highlight URLs that display them, machine learning models can be developed.

Analysis of Website Content: Phishing websites may include misleading material, such as fake login pages or forms intended to collect visitors' personal data. Websites' HTML content and structure can be analyzed by machine learning algorithms to find any suspicious components or resemblances to well-known phishing templates.

Examining an SSL certificate: Phishers frequently utilize SSL certificates to provide the appearance of credibility. These certificates, however, can be flawed or issued by unreliable institutions. To find potential phishing indications, machine learning models can examine SSL certificate parameters like issuer details, expiration dates, and certificate chains.

Website Reputation: Machine learning can assess a website's credibility using reputation ratings and historical data. Models can pinpoint websites connected to prior phishing attempts or other questionable behavior by examining elements like IP reputation, domain age, and hosting location.

Textual Analysis: Phishing attempts frequently use emails or texts to deceive recipients into visiting bogus websites. When analyzing the content of these communications, natural language processing (NLP) tools can be used to spot questionable grammatical constructions, strange language patterns, or urgent requests that are frequently used in phishing scams.

It is necessary to have a labeled dataset that includes both legitimate and phishing websites in order to train machine learning models for phishing website identification. This dataset is used to train the models to spot phishing patterns and telltale signs. This challenge can be completed using a variety of supervised learning methods, including decision trees, random forests, and neural networks.

After being trained, machine learning models can be implemented in real-time systems to assess website features and forecast the possibility of phishing. Websites that have been identified as possible phishing sites can either be prohibited or given more

scrutiny in order to stop users from accessing them. A website that served as a platform for all users inspired our notion. This responsive, interactive website is used to evaluate the reliability of other websites fishing. HTML, CSS, Javascript, Django, and other technologies were used to construct this website. The fundamental framework of a website is built using HTML. To make the website more appealing and user-friendly, it can add effects. Since the website is intended for everyone, using it should be simple and trouble-free for everyone. Even a complete novice should be able to benefit from utilizing this website. On our website, it can find details about the services to provide. It also includes information on unethical conduct in the technological environment of today.

The website was created with the intention of educating visitors about fraudulent actions taking place in today's society and enabling them to identify between legal and fake websitesdebit card numbers, CVVs, bank account details, etc. When assessing if a website URL is legitimate or fake, the dataset contains numerous factors that need to be taken into account are explained in figure 2.

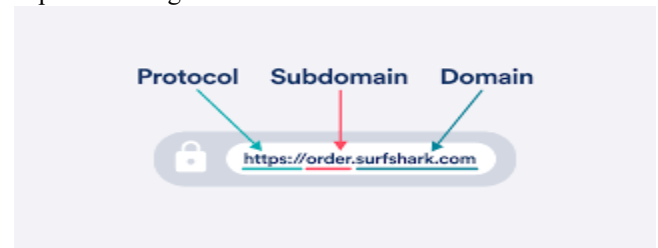


Fig 2: Features of URL

3.1 Elements for phishing website detection and classification

• Address Bar based Features

1. Use of IP Address: If a URL consists of an IP address, such as 125.98.3.123, instead of a domain name, it is highly likely that someone is attempting to obtain personal data from the user.
2. Lengthy URLs with Hidden Suspicious Portions: Scammers may employ long URLs to hide suspicious portions within the address bar, making it difficult for users to identify potential phishing attempts.
3. URL Shortening Services: URLs can be significantly shortened using services like TinyURL, while still directing users to the intended webpage. This can be exploited by attackers to mask malicious URLs.
4. Use of the "@" Sign in URLs: When the "@" sign is used in a URL, the browser ignores

everything before it and often jumps to the actual address after the "@" symbol. Attackers may exploit this to deceive users.

5. URL Redirection: If a URL contains a "/" sign in the route, users may be redirected to a different website. This technique can be utilized by phishers to mislead users and lead them to malicious sites.
6. Added Domains with Hyphens and Prefix/Suffix: Legitimate URLs rarely contain hyphens. Phishers frequently add prefixes or suffixes to domain names, separated by a hyphen, in order to make users believe they are accessing a trusted website.
7. Multiple Domains and Subdomains: Domains can contain multiple levels of subdomains. For example, <http://www.hud.ac.uk/students/>. Phishers may use top-level domains with country codes (ccTLDs) to create deceptive URLs.
8. SSL Protocol (HTTPS): While the presence of HTTPS is an indication of a secure website, it alone is not sufficient to guarantee legitimacy. Phishers can also use SSL certificates to make fraudulent websites appear secure.
9. Duration of Domain Registration: Phishing websites often have short lifespans, mimicking the appearance of legitimate websites. In testing, it was found that many bogus domains had registrations lasting only one year or less.
10. Favicon: A favicon is an icon associated with a specific website. Checking for a legitimate favicon can help verify the authenticity of a website.
11. Port Selection: Choosing different ports can reveal whether a specific service is active on a server, assisting in identifying potential phishing attempts.
12. Deceptive HTTPS Token: Phishers may add the HTTPS token to the domain portion of a URL to deceive users into thinking the website is secure, even if it is not.

These indicators and techniques can assist users in recognizing and avoiding phishing attempts by carefully examining URLs and associated elements.

- ***Abnormal Based Features***

1. External Page Elements in URL: External page elements such as images and videos that

originate from a different domain introduce noise into the URL, making it potentially suspicious.

2. Anchor URL: The anchor component within the <a> tag follows the same guidelines as the request URLs. It is important to ensure that the anchor URLs align with the expected behavior and do not lead to suspicious or phishing websites.

- ***HTML and JavaScript Based Features***

1. Web Page Redirection: The frequency of website hacks blurs the line between trustworthy websites and phishing websites. Websites that have been compromised multiple times raise concerns about their credibility.
2. Disabling Right-Click: Phishers employ JavaScript to disable right-click functionality on their websites, preventing users from accessing the source code or saving it. They may also use onMouseOver methods to conceal the true destination of a link.
3. Pop-Up Windows: Reliable websites rarely employ pop-up windows to solicit personal information. Such requests for personal data are uncommon on legitimate websites and can serve as a red flag for potential phishing attempts.
4. IFrame Redirection: The HTML tag IFrame is used to embed a different webpage within the current one. Phishers may exploit this technique to redirect users to deceptive websites while displaying content from a seemingly legitimate source.

- ***Domain Based Features***

1. Based on our dataset analysis, it has been observed that a majority of phishing websites have a short lifespan. To identify legitimate domains, we have set a criterion that requires DNS records to be at least six months old.
2. A crucial indicator of a phishing website is the absence of an identified host name record or the unavailability of the submitted ID in the WHOIS database. If the DNS record is either blank or non-existent, the website is classified as phishing; otherwise, it is considered legitimate.
3. The Website Traffic feature measures the number of visits and pages viewed on a website, providing insights into its popularity and user engagement.

4. PageRank is a scoring system ranging from 0 to 1 that assesses the significance of a webpage on the Internet. It helps determine the importance and authority of a website.
5. The Google Index function checks whether a web page has been indexed by Google or not. If a website appears in search results, it indicates that Google has included it in its index.
6. The quantity of links pointing to a webpage, including those from the same domain, serves as an indicator of its authority and influence.
7. Statistical-Reports Based Feature: This feature encompasses a range of statistical reports that can provide valuable insights into the characteristics and behavior of a website, aiding in the detection of phishing websites.

These features and indicators contribute to the overall phishing detection process, enabling the identification of potential threats and differentiating between legitimate websites and phishing attempts.

3.2. Algorithms Used

• Random Forest Algorithm

The random forest algorithm uses a number of different elements to construct the forest voting system. A huge tree population yields great detection accuracy. The tree is built using the bootstrap approach. Using samples and record attributes that are chosen at random, the bootstrap approach creates a single tree. From randomly chosen features, the random forest approach chooses the optimal slivers for categorization are explained in figure 3.

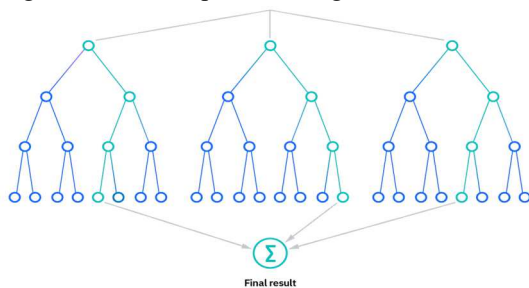


Figure 3: Random Forest

• Decision Tree Algorithm

The best splitter from the available qualities for classification is selected by the decision tree, which is thought of as the tree's root. Algorithm continues to create the tree until it uncovers the leaf node. Each internal node of the tree represents an attribute, and each leaf node has a class label. The decision tree

provides the training model that is used to forecast the target value or class.

IV. RESULTS AND DISCUSSION

The Scikit-learn tool was used to import machine learning techniques. Every classifier undergoes training on the training set, and its effectiveness is assessed on the test set.

1. During the preprocessing stage, nine features are produced from the unstructured data. These traits include the following: the length of the URL, the presence of HTTP, the number of periods and slashes, the inclusion of phishing phrases, the length of the subdomains, and the IP address.
2. A structured data set is then produced. This data collection contains all information pertaining to the couple and is subjected to several classifiers (0,1).
3. Afterwards, they train three different classifiers and assess each one's effectiveness using its accuracy. Two of the classifiers employed are decision trees and random forests.
4. The classifier chooses the supplied URL based on the data it has analyzed. If the website is a phishing site, let the user know that it is a phishing site, and if the website is a real site, let the user know that it is a legitimate site.
5. After comparing the precision of various classifiers, it was discovered that random forest had the best precision.

A brief pop-up notice about this dangerous website will appear on your screen if a user-provided URL is identified as one. The user can use the "Confirm" option to open the website if they need to access specific info on it. Return to the previous page if necessary. It evaluated each classifier's performance using its accuracy score. The accuracy of the decision tree and random forest algorithm are described in figure 4.

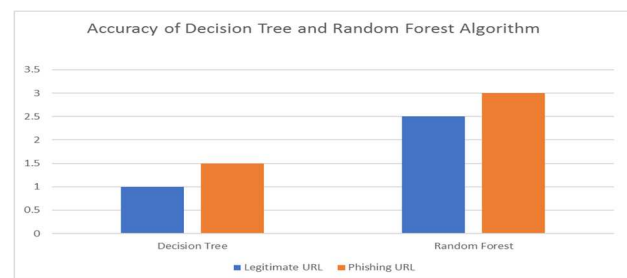


Fig 4: Accuracy of Decision Tree and Random Forest Algorithm

V. CONCLUSION AND FUTURE SCOPE

In conclusion, the study has established the significant risk posed by phishing to internet security and reliability, emphasizing the importance of effective phishing detection. The limitations of commonly used techniques such as blacklisting and heuristic scoring systems have been discussed. To address this challenge, the study evaluated the performance of two machine learning algorithms using the Phishing Website Dataset. Based on the results, the algorithm that achieved the highest accuracy was selected. Subsequently, a Chrome plugin was developed to detect phishing websites, making it easily deployable to end users through browser extensions. With an impressive accuracy rate of 97.31%, the random forest algorithm proved effective in recognizing phishing websites.

However, it is important to note that relying solely on lexical URL cues for detection may not be sufficient, as phishers actively manipulate URLs to evade detection and make their targets unpredictable. Therefore, incorporating additional features, such as the host information, is crucial. The ultimate objective is to create a scalable web service for phishing detection that utilizes online learning to rapidly adapt to new phishing attack patterns and improve model accuracy through advanced feature extraction techniques.

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